

MTL1002: TUTORIAL 3
(LINEAR ALGEBRA)

Notation: $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, $\mathcal{P}_n := \{f \in \mathbb{F}[x] : \deg f < n\}$.

- (1) Find a basis of the row space of the following matrices:

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 2 & 5 \\ 0 & 3 & 1 & 1 \\ 3 & 1 & 0 & 1 \end{pmatrix}.$$

- (2) For $i \in \{1, 2, \dots, m\}$, define the i -th projection $p_i : \mathbb{F}^m \rightarrow \mathbb{F}$ by

$$p_i(x_1, x_2, \dots, x_m) = x_i.$$

- (a) Show that p_i is a linear transformation.
- (b) If $T : \mathbb{F}^m \rightarrow \mathbb{F}$ is a linear transformation then T is an $\mathbb{F}$ -linear combination of the projections, that is, $T = a_1p_1 + a_2p_2 + \dots + a_mp_m$ for some $a_1, \dots, a_m \in \mathbb{F}$.
- (c) Further, show that $S : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is a linear transformation if and only if for each $i \in \{1, 2, \dots, m\}$, the composition $p_i \circ S : \mathbb{F}^n \rightarrow \mathbb{F}$ is a linear transformation.
- (d) If $S : \mathbb{F}^n \rightarrow \mathbb{F}^m$ is a linear transformation then $S(x_1, x_2, \dots, x_n) = (y_1, y_2, \dots, y_m)$ where $y_i = a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n$ for some $a_{ij} \in \mathbb{F}$ with $1 \leq i \leq m, 1 \leq j \leq n$.

- (3) Find the rank and nullity of the following linear transformations. Also, write a basis of the range space in each case.

(a) $T : \mathbb{F}^3 \rightarrow \mathbb{F}^3$ defined by $T(x, y, z) = (x + y + z, \quad x - y + z, \quad x + z)$.

- (b) Assume that $0 \leq m \leq n$. Define $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ by

$$T(x_1, x_2, \dots, x_n) = (x_1, x_2, \dots, x_m).$$

- (4) Write the matrix representations of the linear operators with respect to the given ordered basis $\mathcal{B}$.

(a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(x, y) = (x, y)$ and $\mathcal{B} = \{(1, 1), (1, -1)\}$.

(b) $\mathcal{D} : \mathcal{P}_{n+1} \rightarrow \mathcal{P}_{n+1}$ such that $\mathcal{D}(a_0 + a_1x + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1}$ and $\mathcal{B} = \{1, x, \dots, x^n\}$.

- (5) Suppose $\dim V = \dim W < \infty$ and $T : V \rightarrow W$ is a linear transformation. Show that the following statements are equivalent.

- (a) T is an isomorphism.
- (b) T is injective (i.e., one to one).
- (c) $\ker T = 0$.
- (d) T is surjective (i.e., onto).

- (6) Suppose $m > n$. Justify the following statements.
- (a) There is no one to one (injective) $\mathbb{R}$ -linear transformation from $\mathbb{R}^m$ to $\mathbb{R}^n$.
 - (b) There is no onto (surjective) $\mathbb{R}$ -linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$.
- (7) Find the eigenvalues, eigenvectors and dimension of eigenspaces of the following linear operators.
- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x + y, x)$.
 - (b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (y, x)$.
 - (c) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (y, -x)$.
 - (d) $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by $T(x, y) = (y, -x)$.
 - (e) $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ defined by $T(x_1, x_2, \dots, x_n) = (x_n, x_1, \dots, x_{n-1})$.
 - (f) $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by $T(z_1, z_2) = (z_1 - 2z_2, z_1 + 2z_2)$.
- (8) Suppose $\{\lambda_1, \dots, \lambda_t\}$ is the complete set of distinct eigenvalues of a linear operator $T : V \rightarrow V$. Denote by V_i the eigenspace of λ_i . If B_i is a basis of V_i then show that $\bigcup_{i=1}^t B_i$ is a basis of the sum $\sum_{i=1}^t V_i$. Conclude that it is a direct sum.
- Further, show that T is diagonalizable if and only if $\sum_{i=1}^t V_i = V$. Equivalently, show that T is diagonalizable if and only if $\dim V = \dim V_1 + \dim V_2 + \dots + \dim V_t$.
- (9) Find a basis B such that $[T]_B$ is a diagonal matrix in case T is diagonalizable. Find P such that $[T]_B = P[T]_S P^{-1}$ where S is the standard basis in each case.
- (a) $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by $T(x, y) = (y, -x)$.
 - (b) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (5x - 6y - 6z, -x + 4y + 2z, 3x - 6y - 4z)$.
 - (c) $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by $T(x, y) = (x \cos \theta + y \sin \theta, -x \sin \theta + y \cos \theta)$.
- (10) Characteristic polynomial of a matrix is satisfied by the matrix (Cayley-Hamilton). Use it to find (invertibility and) the inverse of the following linear operators.
- (a) $(x, y, z) \mapsto (x + y + z, x + z, -x + y)$.
 - (b) $(x, y, z) \mapsto (x, x + 2y, x + 2y + 3z)$.
- (11) Let V be a vector space over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. Which of the following map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ is an inner product?
- (a) $\langle (x_1, y_1), (x_2, y_2) \rangle = x_1 x_2 + y_1 y_2 + 3$ for $V = \mathbb{R}^2$ and $\mathbb{F} = \mathbb{R}$.
 - (b) $\langle (x_1, y_1), (x_2, y_2) \rangle = x_1 x_2 - y_1 y_2$ for $V = \mathbb{R}^2$ and $\mathbb{F} = \mathbb{R}$.
 - (c) $\langle (x_1, y_1), (x_2, y_2) \rangle = y_1(x_1 + 2x_2) + y_2(2x_1 + 5x_2)$ for $V = \mathbb{R}^2$ and $\mathbb{F} = \mathbb{R}$.
 - (d) $\langle (x_1, y_1), (x_2, y_2) \rangle = x_1 x_2 + y_1 y_2$ for $V = \mathbb{C}^2$ and $\mathbb{F} = \mathbb{C}$.
 - (e) $\langle (x_1, y_1), (x_2, y_2) \rangle = x_1 \bar{x}_2 - y_1 \bar{y}_2$ for $V = \mathbb{C}^2$ and $\mathbb{F} = \mathbb{C}$.
 - (f) For $V = M_n(\mathbb{C})$ and $\mathbb{F} = \mathbb{C}$, take $\langle A, B \rangle = \text{trace}(A\bar{B})$.
 - (g) For $V = \mathcal{C}[0, 1]$ the space of continuous complex valued functions on the interval $[0, 1]$ and $\mathbb{F} = \mathbb{C}$ take $\langle f, g \rangle := \int_0^1 f(t) \bar{g}(t) dt$ where $f, g \in \mathcal{C}[0, 1]$.

- (12) Suppose $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix} \in M_2(\mathbb{R})$ is such that $a > 0$ and $\det(A) = ad - b^2 > 0$. Show that $\langle X, Y \rangle = X^t A Y$ is an inner product on $\mathbb{R}^2$.
- (13) Suppose V is an inner product space. Define $\|v\| := \sqrt{\langle v, v \rangle}$. Prove the following statements.
- (a) $\|v\| = 0$ if and only if $v = 0$.
 - (b) For $a \in F$, $\|av\| = |a|\|v\|$.
 - (c) $\|v + w\| \leq \|v\| + \|w\|$.
 - (d) $|\|v\| - \|w\|| \leq \|v - w\|$.
 - (e) $\langle v, w \rangle = 0 \iff \|v + w\|^2 = \|v\|^2 + \|w\|^2$.
- (14) Use standard inner product on $\mathbb{R}^2$ over $\mathbb{R}$ to prove the following statement: “A parallelogram is a rhombus if and only if its diagonals are perpendicular to each other.”
- (15) Find with respect to the standard inner product of $\mathbb{R}^3$, an orthonormal basis containing $(1, 1, 1)$.
- (16) Find an orthonormal basis of $\mathcal{P}_3 := \{f(x) \in \mathbb{R}[x] : \deg f(x) < 3\}$ with respect to the inner product defined by $\langle f, g \rangle := \int_0^1 f(t)g(t) dt$.
- (17) Suppose W is a subspace of the finite dimensional inner product space V . Define $W^\perp := \{v \in V : \langle w, v \rangle = 0 \ \forall w \in W\}$. Prove the following statements.
- (a) $W^\perp$ is a subspace of V .
 - (b) $W \cap W^\perp = \{0\}$.
 - (c) $V = W \oplus W^\perp$.
 - (d) $(W^\perp)^\perp = W$.
- (18) Suppose $W = \{(x, y) \in \mathbb{R}^2 : x + y = 0\}$. Find the shortest distance of $(a, b) \in \mathbb{R}^2$ from W with respect to
- (a) the standard inner product,
 - (b) the inner product defined by $\langle (x_1, y_1), (x_2, y_2) \rangle = 2x_1x_2 + y_1y_2$.